CASE

Jordan, 34 · cycling crash · landed on right shoulder · visible step-off over the clavicle · cannot abduct past 60° · pulses intact · **clavicle fracture + AC separation**

BIO 004 · WEEK 2 · CASE DEEP DIVE

The Upper Extremity

A cyclist hits the pavement on her shoulder. The injury walks you through every bone from sternum to fingertip.

PATIENT CASE

Jordan is 34, an experienced road cyclist. On a weekend ride this morning she **swerved to avoid a pothole, lost control, and landed on her right shoulder** at about 20 mph.

She walks into urgent care holding her right arm against her body, supporting it with her left hand. There is a **visible step-off and tenderness over the lateral clavicle**. She cannot **abduct the arm past 60 degrees** without sharp pain. **Distal pulses, sensation, and motor function are intact** in the hand. X-ray shows a **mid-clavicular fracture** and an **acromioclavicular (AC) joint separation**.

Jordan's injury is at the only place the upper limb attaches bony to the trunk. To understand why it changes the way the whole arm moves, you have to walk the upper extremity from where it joins the skeleton all the way to the fingertips, with attention to where bones meet and where nerves and vessels travel.

> GUIDING QUESTION
>
> What bones form the upper extremity, where is it actually attached to the rest of the skeleton, and why does a fracture of the clavicle change how the arm moves?

PREWORK

Companion to your Week 2 Upper Extremity prework page. Terms, video, and spaced recall for the pectoral girdle, arm, forearm, and hand live there.

1 OVERVIEW

Four regions, one chain

The upper extremity has four regions: **pectoral (shoulder) girdle**, **arm** (shoulder to elbow), **forearm** (elbow to wrist), and **hand**. Thirty-two bones per side, plus the joints between them, give the upper limb its remarkable mobility and precision.

Mobility comes at a cost: the upper limb has the loosest, most mobile joints in the body (especially the glenohumeral joint) and the most easily injured attachments. The clavicle is the most fractured bone in the body. The AC joint is the most commonly separated joint. Jordan's injury is a textbook example of the price the upper limb pays for its range of motion.

2 PECTORAL GIRDLE

The only place your arm bony-attaches to your trunk

The pectoral girdle is just two bones per side: the **clavicle** (collarbone) and the **scapula** (shoulder blade). They form three joints between them and the axial skeleton, and they suspend the entire upper limb.

Sternoclavicular (SC) joint

Where the clavicle meets the sternum. *The only bony attachment of the entire upper limb to the axial skeleton.* Saddle joint, very stable, rarely dislocates.

Acromioclavicular (AC) joint

Where the lateral end of the clavicle meets the acromion of the scapula. Plane joint. Commonly separated in falls onto the shoulder. Jordan's.

Glenohumeral joint

Where the head of the humerus meets the glenoid cavity of the scapula. Ball-and-socket. Most mobile joint in the body; least stable.

Scapulothoracic "joint"

Not a true joint — the scapula glides on the posterior thorax via muscle attachments. Provides additional shoulder motion.

Why this matters for Jordan. Smooth abduction past 90° requires synchronized motion at all three real joints (SC, AC, glenohumeral) plus the scapulothoracic glide. A fractured clavicle uncouples that chain. The arm can hang and the elbow and hand still work, but lifting the arm fails because the lever arm of the shoulder girdle is broken.

3 ARM (HUMERUS)

One bone, three nerve relationships worth remembering

The arm contains a single bone, the **humerus**. From proximal to distal:

Head

Round, articulates with the glenoid.

Greater and lesser tubercles

Attachment sites for rotator cuff muscles.

Surgical neck

Common fracture site; close to the axillary nerve and posterior humeral circumflex artery.

Radial groove (mid-shaft)

Spiral groove on the posterior shaft carries the *radial nerve*. Mid-shaft humeral fractures classically injure this nerve, producing wrist drop ("Saturday night palsy").

Medial epicondyle

The *ulnar nerve* wraps around behind it (the "funny bone").

Distal articular surface

Trochlea articulates with the ulna; *capitulum* articulates with the head of the radius. Forms the elbow joint.

4 FORFARM

Two parallel bones, four critical landmarks

The forearm has two bones lying side by side: **radius** (lateral, thumb side) and **ulna** (medial, little finger side). They rotate around each other to pronate and supinate the hand.

Ulna

Longer, posterior. Proximal end: *olecranon* (point of the elbow) and *trochlear notch* (articulates with humerus). Distal end small.

Radius

Shorter, anterior. Proximal: *head* articulates with capitulum. Distal: larger, articulates with the carpus and with the ulna.

Interosseous membrane

Fibrous sheet between radius and ulna. Transmits force from radius (where the hand attaches) to ulna (where the elbow load goes).

Anatomic snuffbox

Triangular depression at the radial base of the thumb, formed by three thumb tendons. Floor: *scaphoid* bone. Tenderness here after a fall = scaphoid fracture until proven otherwise (FOOSH injury).

5 HAND

Twenty-seven bones, three groups

The hand has 27 bones per side, arranged in three groups: 8 carpal bones in the wrist, 5 metacarpals in the palm, 14 phalanges in the fingers (2 in the thumb, 3 in each of the other fingers).

Carpals (8)

Two rows of 4. Proximal row (lateral to medial): *scaphoid, lunate, triquetrum, pisiform*. Distal row: *trapezium, trapezoid, capitate, hamate*. Mnemonic: "Some Lovers Try Positions That They Can't Handle."

Metacarpals (5)

Numbered I (thumb) to V (little finger). Form the palm. Knuckles are the heads of metacarpals.

Phalanges (14)

Thumb: *proximal, distal*. Other fingers: *proximal, middle, distal*.

Clinical: scaphoid

The most commonly fractured carpal bone, usually from a fall on outstretched hand (FOOSH). Blood supply enters distally; proximal pole can necrose if the fracture displaces. Tenderness in the anatomic snuffbox is the bedside clue.

✎ **DRAW ZONE 1** -- *Sketch the pectoral girdle: clavicle, scapula, the three real joints (SC, AC, glenohumeral). Mark Jordan's fracture and AC separation.*

Use a different color for each joint. Show how the AC separation uncouples the chain.

✎ **DRAW ZONE 2** *-- Sketch the entire upper extremity in anatomical position. Label humerus landmarks, radius, ulna, the eight carpals, metacarpals, and phalanges.*

Star the radial groove (radial nerve), the medial epicondyle (ulnar nerve), and the anatomic snuffbox (scaphoid).

RETURNING TO JORDAN

One crash, one chain, every region tested

Jordan landed on the shoulder. The injury sits at the junction between her trunk and her arm, and the consequences ripple distally even though every distal structure is intact.

"*Visible step-off over the clavicle*"

The lateral fragment is being pulled down by the weight of the arm and the trapezius is pulling the medial fragment up. The step is the deformity in between.

"*Cannot abduct past 60°*"

Initial 0-60° of abduction is mostly glenohumeral. Past that, the scapula needs to rotate (scapulohumeral rhythm), which requires an intact clavicle to transmit force from the trapezius and SC joint.

"*AC joint separation*"

Direct fall on the shoulder drives the acromion down while the clavicle stays where the SC joint anchors it. The coracoclavicular and acromioclavicular ligaments tear. Graded by severity (Rockwood I-VI).

"*Distal pulses and sensation intact*"

Good news. The brachial plexus and subclavian vessels run just deep to the middle third of the clavicle. A displaced fracture there can injure them, but Jordan's neurovascular exam is clean.

"*Why not a sling and forget it*"

Mid-shaft clavicle fractures with significant displacement, comminution, or shortening have higher non-union rates if treated non-operatively. Orthopedic referral decides on plate fixation vs sling.

"*Range of motion plan*"

Even with a fracture, early gentle elbow and hand motion prevents stiffness. Shoulder pendulum exercises start once acute pain settles. Full overhead motion may take 8-12 weeks.

Girdle, arm, forearm, hand. Jordan's injury walks you through every region. When you go back to the prework, the SC, AC, and glenohumeral joints will land differently — they are the three real joints she just stress-tested.